

LAB SESSION 09

ASCII & BCD INSTRUCTIONS

1. Write a program to count the number of 1s in a word entered by a user. The count of number is in BCD. Convert the number into decimal number system and display the count on screen.

Source Code:

```

001 TITLE LAB 9 ACTIVITY 1          ;Title of the program
002 .MODEL SMALL                   ;Set the memory model to SMALL
003 .STACK 100H                   ;Set the stack size to 100H
004 .DATA                          ;Declare data segment
005
006 INPUT DB 'ENTER FOUR DIGIT NUMBER $' ;Define a string for displaying a prompt
007 DIG1 DB ?                      ;Declare variable DIG1 for the first digit
008 DIG2 DB ?                      ;Declare variable DIG2 for the second digit
009 DIG3 DB ?                      ;Declare variable DIG3 for the third digit
010 DIG4 DB ?                      ;Declare variable DIG4 for the fourth digit
011
012 .CODE                          ;Start code segment
013 MAIN PROC                      ;Main procedure
014     MOV AX,@DATA                ;Load the address of the data segment into AX
015     MOV DS,AX                  ;Set DS to the value in AX
016
017     MOV BX,0H                  ;Initialize BX register to zero
018
019     MOV DX,OFFSET INPUT          ;Load the effective address of the INPUT string into DX
020     MOV AH,9                   ;Set the function number for displaying a string
021     INT 21H                    ;Call DOS interrupt 21H to display the string
022
023     ;Read the first digit
024     MOV AH,1                   ;Set the function number for reading a character
025     INT 21H                    ;Call DOS interrupt to read a character
026     MOV DIG1,AL                ;Store the first digit in variable DIG1
027
028     ;Read the second digit
029     MOV AH,1                   ;Set the function number for reading a character
030     INT 21H                    ;Call DOS interrupt to read a character
031     MOV DIG2,AL                ;Store the second digit in variable DIG2
032
033     ;Read the third digit
034     MOV AH,1                   ;Set the function number for reading a character
035     INT 21H                    ;Call DOS interrupt to read a character
036     MOV DIG3,AL                ;Store the third digit in variable DIG3
037
038     ;Read the fourth digit
039     MOV AH,1                   ;Set the function number for reading a character
040     INT 21H                    ;Call DOS interrupt to read a character
041     MOV DIG4,AL                ;Store the forth digit in variable DIG4
042
043     MOV AL,DIG1                ;Move the value of DIG1 to AL
044     MOV CX,4                   ;Set the loop counter CX to 4
045     SBACK1:                     ;Start of loop to iterate through each bit of DIG1
046     ROR AL,1                   ;Rotate the bits in AL to the right (right shift)
047     JC COUNT1                  ;Jump to COUNT1 if the carry flag is set
048     LBACK1:                     ;Label LBACK
049     LOOP SBACK1                ;Decrease the CX and continue the loop until CX becomes zero
050     JMP NEXT1                  ;Jump to NEXT1 after the loop
051     COUNT1:                     ;Label COUNT1
052     INC BX                     ;Increment in BX
053     JMP LBACK1                 ;Jump back to the loop
054

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055 NEXT1:                ;Jump to NEXT1
056 MOV AL,DIG2           ;Move the value of DIG2 to AL
057 MOV CX,4              ;Set the loop counter CX to 4
058 SBACK2:               ;Start of loop to iterate through each bit of DIG2
059 ROR AL,1              ;Rotate the bits in AL to the right (right shift)
060 JC COUNT2             ;Jump to COUNT2 if the carry flag is set
061 LBACK2:               ;Label LBACK2
062 LOOP SBACK2           ;Decrease the CX and continue the loop until CX becomes zero
063 JMP NEXT2:            ;Jump to NEXT2 after the loop
064 COUNT2:              ;Label COUNT2
065 INC BX               ;Increment in BX
066 JMP LBACK2           ;Jump back to the loop
067
068 NEXT2:                ;Jump to NEXT2
069 MOV AL,DIG3           ;Move the value of DIG3 to AL
070 MOV CX,4              ;Set the loop counter CX to 4
071 SBACK3:               ;Start of loop to iterate through each bit of DIG3
072 ROR AL,1              ;Rotate the bits in AL to the right (right shift)
073 JC COUNT3            ;Jump to COUNT3 if the carry flag is set
074 LBACK3:               ;Label LBACK3
075 LOOP SBACK3           ;Decrease the CX and continue the loop until CX becomes zero
076 JMP NEXT3            ;Jump to NEXT3 after the loop
077 COUNT3:              ;Label COUNT3
078 INC BX               ;Increment in BX
079 JMP LBACK3           ;Jump back to the loop
080
081 NEXT3:                ;Jump to NEXT3
082 MOV AL,DIG4           ;Move the value of DIG4 to AL
083 MOV CX,4              ;Set the loop counter CX to 4
084 SBACK4:               ;Start of loop to iterate through each bit of DIG4
085 ROR AL,1              ;Rotate the bits in AL to the right (right shift)
086 JC COUNT4            ;Jump to COUNT4 if the carry flag is set
087 LBACK4:               ;Label LBACK4
088 LOOP SBACK4           ;Decrease the CX and continue the loop until CX becomes zero
089 JMP GO                ;Jump to the GO after the loop
090 COUNT4:              ;Label COUNT4
091 INC BX               ;Increment in BX
092 JMP LBACK4           ;Jump back to the loop
093
094 GO:                   ;Label GO
095 MOV AL,BX             ;Move the value in BX to AL
096 MOV DL,01             ;Move the value 01 to DL
097 MUL DL               ;Multiply AL by DL, result in AX
098 AAM                  ;ASCII Adjust for Multiplication
099 OR AX,3030H           ;Convert the result to ASCII in AX
100 MOV BX,AX            ;Move the ASCII result to BX
101
102 MOV AH,2              ;Set the function number for displaying a character
103 MOV DL,0DH            ;Set DL to the ASCII value for carriage return (CR)
104 INT 21H               ;Call DOS interrupt to display the character
105 MOV DL,0AH            ;Set DL to the ASCII value for line feed
106 INT 21H               ;Call DOS interrupt to display the character
107
108 MOV AH,2              ;Set AH to 2 for displaying a character
109 MOV DL,BX             ;Move the result to DL
110 INT 21H               ;Call DOS interrupt to display the character in DL
111
112 MOV AH,2              ;Set the function number for displaying a character
113 MOV DL,BL             ;Move the result to DL
114 INT 21H               ;Call DOS interrupt to display the character
115
116 MOV AH,4CH            ;Set the function number for program termination
117 INT 21H               ;Call DOS interrupt to terminate the program
118
119 MAIN ENDP             ;End of main procedure
120 END MAIN              ;End of the main program

```

OUTPUT:

 emulator screen (80x25 chars)

ENTER FOUR DIGIT NUMBER 0175
06

In BCD codes:

0 = 0000,

1 = 0001,

7 = 0111,

5 = 0101,

total number of 1s = 6

- Take two 1 digit inputs (from 0 to 9) from user through keyboard. Write a program to perform the following operations, and display the result for all the operation on screen.

- Addition**

Source Code:

```

01 TITLE LAB 9 ACTIVITY 2A      ;Title of the program
02 .MODEL SMALL                ;Define the memory model as SMALL
03 .STACK 100H                 ;Set the stack size to 100H (256 bytes)
04
05 .DATA                       ;Initialize Data Segment
06 FIRST_NUMBER DB 'ENTER FIRST NUMBER: $' ;Define a string for the first number prompt
07 SECOND_NUMBER DB 0DH, 0AH, 'ENTER SECOND NUMBER: $' ;Define a string for the second number prompt
08 RESULT DB 0DH, 0AH, 'THE RESULT IS = $' ;Define a string for the result message
09
10 .CODE                       ;Initialize Code Segment
11 MAIN PROC                   ;Start of the main procedure
12     MOV AX, @DATA            ;Load the data segment address into AX
13     MOV DS, AX               ;Set the data segment register to the data segment address
14
15     LEA DX, FIRST_NUMBER     ;Load the effective address of the first number prompt into DX
16     MOV AH, 9                ;Set AH to 9 for the "print string" function
17     INT 21H                  ;Call DOS interrupt 21H to print the first number prompt
18
19     MOV AH, 1                ;Set AH to 1 for the "read character" function
20     INT 21H                  ;Call DOS interrupt 21H to read a character
21     MOV BL, AL               ;Move the input character to BL register
22
23     LEA DX, SECOND_NUMBER    ;Load the effective address of the second number prompt into DX
24     MOV AH, 9                ;Set AH to 9 for the "print string" function
25     INT 21H                  ;Call DOS interrupt 21H to print the second number prompt
26
27     MOV AH, 1                ;Set AH to 1 for the "read character" function
28     INT 21H                  ;Call DOS interrupt 21H to read a character
29
30     MOV AH, 00H              ;Set AH to 0 for the "add" function
31     XCHG AL, BL              ;Exchange the values in AL and BL
32     ADD AL, BL               ;Add the values in AL and BL
33     AAA                     ;ASCII adjust after addition
34
35     OR AX, 3030H             ;Convert the result to ASCII
36     MOV BX, AX               ;Move the ASCII result to BX register
37
38     LEA DX, RESULT           ;Load the effective address of the result message into DX
39     MOV AH, 9                ;Set AH to 9 for the "print string" function
40     INT 21H                  ;Call DOS interrupt 21H to print the result message
41
42     MOV AH, 2                ;Set AH to 2 for the "write character" function
43     MOV DL, BH               ;Move the high digit of the result to DL
44     INT 21H                  ;Call DOS interrupt 21H to write the high digit
45
46     MOV AH, 2                ;Set AH to 2 for the "write character" function
47     MOV DL, BL               ;Move the low digit of the result to DL
48     INT 21H                  ;Call DOS interrupt 21H to write the low digit
49
50     MOV AH, 4CH              ;Set AH to 4CH for the "terminate with return code" function
51     INT 21H                  ;Call DOS interrupt 21H to terminate the program
52
53 MAIN ENDP                   ;End of the main procedure
54 END MAIN                     ;End of the program

```

OUTPUT:

 emulator screen (80x25 chars)

```

ENTER FIRST NUMBER: 3
ENTER SECOND NUMBER: 4
THE RESULT IS = 07

```

• Subtraction

Source Code:

```

01 TITLE LAB 9 ACTIVITY 2B           ;Title of the program
02 .MODEL SMALL                     ;Set the memory model to small
03 .STACK 100H                      ;Set the stack size to 100H bytes
04
05 .DATA                             ;Initialize Data segment
06 FIRST_NUMBER DB 'ENTER FIRST NUMBER: $' ;Define a string for the first prompt
07 SECOND_NUMBER DB 0DH, 0AH, 'ENTER SECOND NUMBER: $' ;Define a string for the second prompt
08 RESULT DB 0DH, 0AH, 'THE DIFFERENCE IS = $' ;Define a string for the result message
09
10 .CODE                             ;Initialize Code segment
11 MAIN PROC                         ;Start of the main procedure
12     MOV AX,@DATA                  ;Load the address of the data segment into AX
13     MOV DS,AX                    ;Set the data segment register to the loaded address
14
15     LEA DX,FIRST_NUMBER           ;Load the effective address of the first prompt into DX
16     MOV AH,9                     ;Set AH to 9 for the "print string" function
17     INT 21H                      ;Call interrupt 21H to print the first prompt
18
19     MOV AH,1                     ;Set AH to 1 for the "read character" function
20     INT 21H                      ;Call interrupt 21H to read a character from the user
21     MOV BL,AL                    ;Move the input character to BL
22
23     LEA DX,SECOND_NUMBER          ;Load the effective address of the second prompt into DX
24     MOV AH,9                     ;Set AH to 9 for the "print string" function
25     INT 21H                      ;Call interrupt 21H to print the second prompt
26
27     MOV AH,1                     ;Set AH to 1 for the "read character" function
28     INT 21H                      ;Call interrupt 21H to read a character from the user
29
30     MOV AH,00H                   ;Set AH to 00H
31     XCHG AL,BL                    ;Exchange the values in AL and BL
32     SUB AL,BL                     ;Subtract the value in BL from AL
33     AAS                           ;Adjust after subtraction
34     OR AX,3030H                  ;Convert the result to ASCII
35     MOV BX,AX                    ;Move the ASCII result to BX
36
37     LEA DX,RESULT                 ;Load the effective address of the result message into DX
38     MOV AH,9                     ;Set AH to 9 for the "print string" function
39     INT 21H                      ;Call interrupt 21H to print the result message
40
41     MOV AH,2                     ;Set AH to 2 for the "print character" function
42     MOV DL,BH                     ;Move the high digit of the result to DL
43     INT 21H                      ;Call interrupt 21H to print the high digit
44
45     MOV AH,2                     ;Set AH to 2 for the "print character" function
46     MOV DL,BL                     ;Move the low digit of the result to DL
47     INT 21H                      ;Call interrupt 21H to print the low digit
48
49     MOV AH,4CH                   ;Set AH to 4CH for the "terminate with return code" function
50     INT 21H                      ;Call interrupt 21H to terminate the program
51
52 MAIN ENDP                         ;End of the main procedure
53 END MAIN                          ;End the program
54

```

OUTPUT:

 emulator screen (80x25 chars)

```

ENTER FIRST NUMBER: 9
ENTER SECOND NUMBER: 2
THE DIFFERENCE IS = 07

```

• Multiplication

Source Code:

```

01 TITLE LAB 9 ACTIVITY 2C      ;Title of the program
02 .MODEL SMALL                ;Define memory model as SMALL
03 .STACK 100H                 ;Set the stack size to 100H bytes
04 .DATA                       ;Initialize Data segment
05 FIRST_NUMBER DB 'ENTER FIRST NUMBER: $' ;String for displaying prompt
06 SECOND_NUMBER DB 0DH, 0AH, 'ENTER SECOND NUMBER: $' ;String for displaying prompt
07 RESULT DB 0DH, 0AH, 'THE RESULT IS = $' ;String for displaying result
08
09 .CODE                       ;Initilize Code segment
10 MAIN PROC                   ;Start of the main procedure
11     MOV AX,@DATA             ;Load the address of the data segment into AX
12     MOV DS,AX                ;Set the data segment register to the value in AX
13
14     LEA DX,FIRST_NUMBER      ;Load the effective address of the FIRST_NUMBER string into DX
15     MOV AH,9                 ;Set AH to 9 to indicate print string function
16     INT 21H                  ;Call interrupt 21H to print the string
17
18     MOV AH,1                 ;Set AH to 1 to indicate read character function
19     INT 21H                  ;Call interrupt 21H to read a character
20     AND AL,0FH               ;Mask the value to 15
21     MOV BL,AL                ;Move the value to BL register
22
23     LEA DX,SECOND_NUMBER     ;Load the effective address of the SECOND_NUMBER string into DX
24     MOV AH,9                 ;Set AH to 9 to indicate print string function
25     INT 21H                  ;Call interrupt 21H to print the string
26
27     MOV AH,1                 ;Set AH to 1 to indicate read character function
28     INT 21H                  ;Call interrupt 21H to read a character
29     MOV AH,00H               ;Clear AH to prepare for multiplication
30     AND AL,0FH               ;Mask the value to 15
31     MUL BL                   ;Multiply AL by BL, result in AX
32     AAM                      ;ASCII adjust for multiplication
33     OR AX,3030H              ;Convert the result to ASCII by ORing with '30' in hex
34     MOV BX,AX                ;Move the result to BX register
35
36
37     LEA DX,RESULT            ;Load the effective address of the RESULT string into DX
38     MOV AH,9                 ;Set AH to 9 to indicate print string function
39     INT 21H                  ;Call interrupt 21H to print the string
40
41
42     MOV AH,2                 ;Set AH to 2 to indicate print character function
43     MOV DL,BH                ;Load the tens digit into DL
44     INT 21H                  ;Call interrupt 21H to print the character
45
46     MOV AH,2                 ;Set AH to 2 to indicate print character function
47     MOV DL,BL                ;Load the units digit into DL
48     INT 21H                  ;Call interrupt 21H to print the character
49
50     MOV AH,4CH               ;Set AH to 4CH to indicate program termination function
51     INT 21H                  ;Call interrupt 21H to terminate the program
52
53 MAIN ENDP                   ;End of the main procedure
54 END MAIN                     ;End of the program

```

OUTPUT:

 emulator screen (80x25 chars)

```

ENTER FIRST NUMBER: 7
ENTER SECOND NUMBER: 7
THE RESULT IS = 49

```


• Division

Source Code:

```

01 TITLE LAB 9 ACTIVITY 2D                ;Title of the program
02 .MODEL SMALL                          ;Set the memory model to SMALL
03 .STACK 100H                           ;Set the stack size to 100H
04 .DATA                                 ;Declare data segment
05
06 FIRST_NUMBER DB 'ENTER FIRST NUMBER : $' ;Define a string for displaying a prompt
07 SECOND_NUMBER DB 0DH, 0AH, 'ENTER SECOND NUMBER : $' ;Define a string for displaying a prompt
08 STR1 DB 0DH, 0AH, 'REMAINDER IS = $' ;Define a string for displaying the remainder
09 STR2 DB 0DH, 0AH, 'QUOTIENT IS = $' ;Define a string for displaying the quotient
10
11 .CODE                                ;Start code segment
12 MAIN PROC                            ;Main procedure
13     MOV AX,@DATA                      ;Load the address of the data segment into AX
14     MOV DS,AX                         ;Set DS to the value in AX
15
16     LEA DX,FIRST_NUMBER               ;Load effective address of the FIRST_NUMBER string into DX
17     MOV AH,9                          ;Set the function 9 for displaying a string
18     INT 21H                           ;Call DOS interrupt to display the string
19
20     MOV AH,1                          ;Set the function 1 for reading a character
21     INT 21H                           ;Call DOS interrupt to read a character
22     AND AL,0FH                        ;Mask upper bits, keeping only the lower 4 bits
23     MOV BL,AL                          ;Move the result to register BL
24
25     LEA DX,SECOND_NUMBER               ;Load effective address of the SECOND_NUMBER string into DX
26     MOV AH,9                          ;Set the function 9 for displaying a string
27     INT 21H                           ;Call DOS interrupt to display the string
28
29     MOV AH,1                          ;Set the function 1 for reading a character
30     INT 21H                           ;Call DOS interrupt to read a character
31
32     MOV AH,00H                        ;Clear AH register
33     AND AL,0FH                        ;The four bits of AL, keeping only the lower 4 bits
34     XCHG AL,BL                        ;Exchange the values in AL and BL
35     AAD                               ;ASCII Adjust for Division
36     DIV BL                            ;Divide AX by the value in BL
37     OR AX,3030H                       ;Convert the result to ASCII in AX
38     MOV BX,AX                         ;Move the ASCII result to BX
39
40     LEA DX,STR2                        ;Load effective address of the STR2 string into DX
41     MOV AH,9                          ;Set the function number for displaying a string
42     INT 21H                           ;Call DOS interrupt to display the string
43
44     MOV AH,2                          ;Set the function number for displaying a character
45     MOV DL,BL                          ;Move the quotient to DL
46     INT 21H                           ;Call DOS interrupt to display the character
47
48     LEA DX,STR1                        ;Load effective address of the STR1 string into DX
49     MOV AH,9                          ;Set the function number for displaying a string
50     INT 21H                           ;Call DOS interrupt to display the string
51
52     MOV AH,2                          ;Set the function number for displaying a character
53     MOV DL,BH                          ;Move the remainder to DL
54     INT 21H                           ;Call DOS interrupt to display the character
55
56     MOV AH,4CH                        ;Set the function number for program termination
57     INT 21H                           ;Call DOS interrupt to terminate the program
58
59 MAIN ENDP                            ;End of main procedure
60 END MAIN                             ;End of the main program

```

OUTPUT:

 emulator screen (80x25 chars)

```

ENTER FIRST NUMBER: 8
ENTER SECOND NUMBER: 2
QUOTIENT IS = 4
REMAINDER IS = 0

```

3. Write a program to make a counter that counts up from 0 to 99 in BCD and display the result line by line.

Source Code:

```

01 TITLE LAB 9 ACTIVITY 3 ;Title of the program
02 .MODEL SMALL ;Declare the memory model as SMALL
03 .STACK 100H ;Set the stack size to 100H
04
05 .CODE ;Start of code segment
06
07 MAIN PROC ;Start of the main procedure
08     MOV BL,0 ;Initialize BL register to 0
09     AGAIN: ;Label for the loop
10     MOV DL,1 ;Move the value 1 into DL register
11     MOV AL,BL ;Move the value of BL into AL register
12
13     MUL DL ;Multiply the contents of AL by DL
14     AAM ;ASCII Adjust for Multiplication
15     OR AX,3030H ;Perform bitwise OR operation with 3030H to convert the result to ASCII
16     MOV CX,AX ;Move the ASCII result to CX for display
17
18     MOV AH,2 ;Set the DOS function number 2, which is used for character output
19     MOV DL,CH ;Move the high nibble of CX to DL
20     INT 21H ;Call DOS interrupt 21H to output the first digit
21
22     MOV DL,CL ;Move the low nibble of CX (second digit in ASCII) to DL
23     INT 21H ;Call DOS interrupt 21H to output the second digit
24
25     MOV AH,2 ;Set the DOS function number 2, which is used for character output
26     MOV DL,0AH ;Move the ASCII code for newline into the DL register
27     INT 21H ;Call DOS interrupt 21H to perform newline character output
28
29     MOV DL,0DH ;Move the ASCII code for carriage return into the DL register
30     INT 21H ;Call DOS interrupt 21H to perform carriage return character output
31
32
33     INC BL ;Increment the value in the BL register
34     CMP BL,100 ;Compare the value in BL with 100
35     JNE AGAIN ;Jump to the AGAIN label if the values are not equal
36
37     MOV AH,4CH ;Set the DOS function number 4CH, which is used to exit the program
38     INT 21H ;Call DOS interrupt 21H to terminate the program
39
40     MAIN ENDP ;End of the main procedure
41 END MAIN ;End of the program

```

OUTPUT:

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```

4. Write a program to ask user to enter his/her name in lower case, convert them in upper case, and display the name on screen. Use buffered input service "0AH" of "INT 21H" to take input from user.

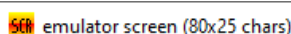
Source Code:

```

01 TITLE LAB 9 ACTIVITY 4          ;Title of the program
02 .MODEL SMALL                   ;Set the memory model to SMALL
03 .STACK 100H                    ;Set the stack size to 100H
04 .DATA                          ;Declare the data segment
05 FIRSTNAME DB 'ENTER FIRST NAME IN LOWER CASE : $' ;String to prompt for input
06 INPUTNAME DB 10, ?, 10 DUP (?) ;Buffer for input name
07 UPPERCASE DB 8 DUP ('$')        ;Buffer to store the uppercase version of the input name
08 DISPLAY DB 'YOUR FIRST NAME IS: $' ;String to display the result
09
10 .CODE                          ;Start of the code segment
11 MAIN PROC                      ;Start of the main procedure
12     MOV AX,@DATA                ;Load the address of the data segment into AX
13     MOV DS,AX                  ;Set DS to the address in AX
14
15     MOV AH,9                    ;Set up to display a string
16     LEA DX,FIRSTNAME            ;Load the offset address of the FIRSTNAME string into DX
17     INT 21H                     ;Call interrupt 21H to display the prompt
18
19     MOV AH,0AH                  ;Set up to input a string
20     MOV DX,OFFSET INPUTNAME      ;Load the offset address of the INPUTNAME buffer into DX
21     INT 21H                     ;Call interrupt 21H to input a string
22
23     MOV SI,OFFSET INPUTNAME      ;Load SI with the offset address of the input string
24     MOV BX,OFFSET UPPERCASE      ;Load BX with the offset address of the uppercase buffer
25
26     MOV CX,11                   ;Set the loop counter to 11
27     AGAIN:                      ;
28     MOV AL,[SI]+2                ;Load the ASCII value of the character from the input string into AL
29     CMP AL,61H                  ;Compare AL with 'a' (61H)
30     JB OVER                      ;Jump to OVER if AL is below 'a'
31     CMP AL,7AH                  ;Compare AL with 'z' (7AH)
32     JA OVER                      ;Jump to OVER if AL is above 'z'
33     AND AL,11011111B            ;Convert the lowercase character to uppercase
34     OVER:                      ;
35     MOV [BX],AL                  ;Store the modified character in the UPPERCASE buffer
36     INC SI                      ;Move to the next character in the input string
37     INC BX                      ;Move to the next position in the UPPERCASE buffer
38     LOOP AGAIN                  ;Repeat the loop until CX becomes zero
39
40     MOV AH,2                    ;Set up to display a new line character
41     MOV DL,0AH                  ;New line character
42     INT 21H                     ;Call interrupt 21H to display a new line
43
44     MOV DL,0DH                  ;Carriage return character
45     INT 21H                     ;Call interrupt 21H to display a carriage return
46
47     MOV AH,9                    ;Set up to display a string
48     LEA DX,DISPLAY              ;Load the offset address of the DISPLAY string into DX
49     INT 21H                     ;Call interrupt 21H to display the result
50
51     MOV AH,9                    ;Set up to display a string
52     MOV DX,OFFSET UPPERCASE      ;Load the offset address of the UPPERCASE buffer into DX
53     INT 21H                     ;Call interrupt 21H to display the uppercase version
54
55     MOV AH,4CH                  ;Set up to terminate the program
56     INT 21H                     ;Call interrupt 21H to terminate the program
57
58     MAIN ENDP                  ;End of the main procedure
59 END MAIN                       ;End of the program

```

OUTPUT:

 emulator screen (80x25 chars)

```

ENTER FIRST NAME IN LOWER CASE: shahzaib
SR FIRST NAME IS: SHAHZAIB

```